

lec

Unsupervised Learning

Data $D = \{x_i\}_{i=1}^n \sim \text{iid } \mathbb{P}_x$

Goal: Estimate $\mathbb{P}_x$ given D .

Model $\mathbb{P}_x$ via a latent variable distri.

$$p_\theta(x) = \sum_z p_\theta(x, z)$$

z : latent variable.

Once $p_\theta(x)$ is estimate, one can estimate $p_\theta(z|x)$: clustering x , if z is disc
: Emb / feature extract if z is continuous.

Case 1: Here z is discrete

$$\exists x \in \mathbb{R}^d$$

$$p_\theta(x) = \sum_z p_\theta(z) \cdot p_\theta(x|z)$$

$$= \sum_{m=1}^M \alpha_m \cdot \mathcal{N}(x; \mu_m, \Sigma_m)$$

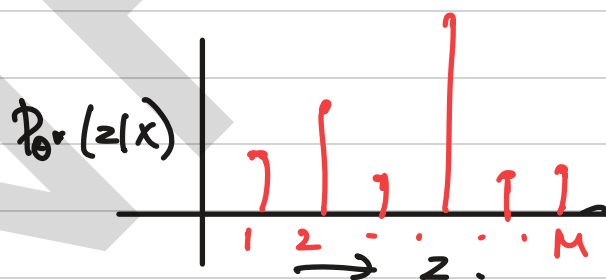
Estimate $\theta^* = \{\alpha^*, \mu^*, \Sigma^*\}$ via EM.

Having estimated θ^* , given a datapoint $x \sim p_x$, we estimate / calculate

$$p_{\theta^*}(z|x) = \frac{p_{\theta^*}(x|z) \cdot p_{\theta^*}(z)}{p_{\theta^*}(x)}$$

: distribution over $z|x$.

for $\forall x \sim p_x$,



↳ represents 'soft-clustering'

GMM-based clustering.

Suppose we want $p_{\theta^*}(z|x)$ to be dirac deltas.

Constituent Gaussians

Lt $\mathcal{N}(x; \mu_m, \sigma^2 I)$ with this,
 $\sigma \rightarrow 0$

$\phi_\theta(z|x)$ will be dirac-deltas.

Recall EM algo:

E-step: Compute $\mathbb{E}_{q_{z|x} = \phi_\theta(z|x)} \log \phi_\theta(x|z)$

M-step: $\arg \max_{\theta} \left[\downarrow \right]$

M-means Clustering Algo:

Initialize M-mean vectors randomly

While convergence do:

i) Assign each x_i to one of the

M-means via $\arg \min_{1 \leq m \leq M} \|x_i - \mu_m\|_2^2$

ii) recalculate μ_m , after the assignment as

$$\mu_j = \frac{1}{|S_j|} \sum_{x_i \in S_j} x_i \quad \text{where}$$

S_j is the set of all datapoints
that got assigned to μ_j in step (i).

End while upon Convergence

Implicit latent variable Models.

p_x is estimated via an implicit
lat. variable model.

Example:

i) linear model on data

$$x \in \mathbb{R}^d, \quad \mathcal{D} = \{x_1, \dots, x_n\}$$

$$z \in \mathbb{R}^k, \quad k < d.$$

$$z = Wx \quad W \in \mathbb{R}^{k \times d}$$

$k \times d \quad d \times 1$

Question : How to obtain W s.t
a "loss function" is optimized.

Define the loss function as, that
of maximizing projected variance.

$$\frac{1}{n} \sum_{i=1}^n \left\| \left[z_i - \frac{1}{n} \sum_{i=1}^n z_i \right] \right\|^2 = L(w, x)$$

$$W^* = \operatorname{argmax}_W L(w, x)$$

$$L(w, x) = \frac{1}{n} \sum \left\| \left[Wx_i - \frac{1}{n} \sum_{i=1}^n Wx_i \right] \right\|^2$$

$$\text{Let } \hat{x} = \frac{1}{n} \sum_{i=1}^n x_i$$

$$\hat{\Sigma} = \frac{1}{n} \sum_{i=1}^n (x_i - \hat{x})(x_i - \hat{x})^T$$

$$L(w, x) = W^T \hat{\Sigma} W$$

$$W^* = \operatorname{argmax}_W W^T \hat{\Sigma} W$$

s.t $W^T W = I$

$$\text{Lag} : W^T \hat{\Sigma} W + \lambda W^T W$$

W^* = Σ vector of $\hat{\Sigma}$ corresponding to the largest Eigen value.

$W_{K \times d}$ matrix will contain the Eigen vectors of $\hat{\Sigma}$, rank-ordered in accordance with their Σ .vals.

Principal Component Analysis (PCA).

$$Z = DNN(x)$$

$$p_0(x) p_0(z|x) = p_0(x, z)$$

$$\sum p_0(x) \cdot p_0(z|x) = p_0(x).$$

Implementing the above Marginalization.

$$x \xrightarrow[\text{Enc.}]{E_{\phi}(x)} z \xrightarrow[\text{Dec.}]{D_{\theta}(z)} \hat{x}$$

$$\hat{x} = D_{\theta}[E_{\phi}(x)]$$

$$L_{011}(\theta, \phi) = \frac{1}{n} \sum_{i=1}^n \|x_i - \hat{x}_i\|_2^2$$

Auto-Encoder.

